Number Theory: Integer Representations and Algorithms

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Section Summary

- Integer Representations
  - Base *b* Expansions
  - Binary Expansions
  - Octal Expansions
  - Hexadecimal Expansions
- Base Conversion Algorithm
- Algorithms for Integer Operations
  - Operations with binary representations

Presentations of Integers

- Integer can be represented using any base *b*, $\boldsymbol{b \in \mathbb{Z}^{+} \; and \; b > 1}$
  - Possible values of a digit in different bases
    - In base 10 expansions: values are $\{0,1,2,3,4,5,6,7,8,9\}$
    - In base 2 expansions: values are $\{0,1\}$
    - In base 8 expansions: values are $\{0,1,2,3,4,5,6,7\}$
    - In base 16 expansions: values are $\{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F\}$
  - These bases are the ones commonly used in Computer Science

Understand the Base$_1$

- Try binary representation
  - 0000 is the first
  - 0001 is the second -- there is 1 smaller number
  - 0010 is the third -- there are 2 smaller numbers
  - 0011 is the fourth
  - 0100 is the fifth -- there are 4 smaller numbers
  - 0101 is the sixth
  - 0110 is the seventh
  - 0111 is the eighth
  - 1000 is the ninth -- there are 8 smaller numbers
    - The weight of digit i is from an exponential function
    - From binary to decimal is a one-to-one mapping

Increase by 1: when there will be a carry?

How many different numbers can be represented by n digits?

Understand the Base$_2$

- For each digit there are two possible values
  - Using $(1011)_2$ as an example
  - $(1011)_2 = (1000)_2 + (10)_2 + (1)_2$
    - No carry bit in this example, so this summation is easy

Binary	**1**	**0**	**1**	**1**
N-th Digit	4	3	2	1
# Smaller (or **Weight**)	$\mathbf{8 = 2^3}$	$\mathbf{4 = 2^2}$	$\mathbf{2 = 2^1}$	$\mathbf{1 = 2^0}$
Base 10 (sum from right)	$1 * 2^3 +$ $0 * 2^2 +$ $1 * 2^1 +$ $1 * 2^0$	$0 * 2^2 +$ $1 * 2^1 +$ $1 * 2^0$	$1 * 2^1 +$ $1 * 2^0$	$1 * 2^0$

- $\sum_{i=0}^{k-1} 2^i < 2^k$: from decimal to binary is one-to-one mapping

Base b Representations

- **Theorem 1**: For any $b \in \mathbb{Z}^+ \wedge b > 1$ and $n \in \mathbb{Z}^+$, n can be expressed uniquely in the form:

$$\boldsymbol{n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b^1 + a_0 b^0}$$

  where $k \in \mathbb{N}$, $a_i \in \mathbb{N} \wedge a_i < b$, for $i = 0,1,2,\ldots,k$, and $a_k \neq 0$.
- Base b expansion of n
- a_0, a_1, …, a_k are the base-*b* digits
- n can be denoted as $(a_k a_{k-1} \ldots a_1 a_0)_b$

Binary Expansions

- **Example**: What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?
- **Solution**: $(1\ 0101\ 1111)_2 = 1\cdot 2^8 + 0\cdot 2^7 + 1\cdot 2^6 + 0\cdot 2^5 + 1\cdot 2^4 + 1\cdot 2^3 + 1\cdot 2^2 + 1\cdot 2^1 + 1\cdot 2^0 = 351.$

- **Example**: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?
- **Solution**: $(11011)_2 = 1\cdot 2^4 + 1\cdot 2^3 + 0\cdot 2^2 + 1\cdot 2^1 + 1\cdot 2^0 = 27.$

Octal Expansions

- **Example**: What is the decimal expansion of the number with octal expansion $(7016)_8$?
- **Solution**: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

- **Example**: What is the decimal expansion of the number with octal expansion $(111)_8$?
- **Solution**: $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

Hexadecimal Expansions

- **Example**: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?
- **Solution**: $2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$

- **Example**: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?
- **Solution**: $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$

Base Conversion$_0$

- n can be denoted as $(a_k a_{k-1} \dots a_1 a_0)_b$

$$\boldsymbol{n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b^1 + a_0 b^0}$$

- How to construct the base *b* expansion of an integer *n ?*
  - How to get $\boldsymbol{a_0}$? n % b
  - How to get $\boldsymbol{a_1}$? (n / b) % b
  - How to get $\boldsymbol{a_2}$? (n / b^2) % b
  - ….
  - n can be denoted as $(a_k a_{k-1} \dots a_1 a_0)_b$

Base Conversion$_1$

- To construct the base *b* expansion of an integer *n*:
  - Divide *n* by *b* to obtain a quotient and remainder.

$$n = bq_0 + a_0 \quad 0 \leq a_0 \leq b$$

  - The remainder, a_0, is the rightmost digit in the base *b* expansion of *n*. Next, divide q_0 by *b*.

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 \leq b$$

  - The remainder, a_1, is the second digit from the right in the base *b* expansion of *n*.
  - Continue by successively dividing the quotients by *b*, obtaining the additional base *b* digits as the remainder. The process terminates when the quotient is 0.

Base Conversion$_2$

- Algorithm 1: constructing base b expansions

procedure *base b expansion*(n, b: positive integers with $b > 1$)
  $q := n$
  $k := 0$
  while ($q \neq 0$)
    $a_k := q$ **mod** b
    $q := q$ **div** b
    $k := k + 1$
return($a_{k-1}, \ldots, a_1, a_0$){$(a_{k-1} \ldots a_1 a_0)_b$ is base b expansion of n}

- q represents the quotient obtained by successive divisions by b, starting with $q = n$.
- The digits in the base b expansion are the remainders of the division, given by q **mod** b.

Base Conversion$_3$

- **Example**: Find the octal expansion of $(12345)_{10}$
- **Solution**: Successively dividing by 8 gives:
  - $12345 = 8 \cdot 1543 + 1$
  - $1543 = 8 \cdot 192 + 7$
  - $192 = 8 \cdot 24 + 0$
  - $24 = 8 \cdot 3 + 0$
  - $3 = 8 \cdot 0 + 3$
- The remainders are the digits from right to left yielding $(30071)_8$.

Compare with decimals: how to get each digit of a decimal number? The same thing: modular + division

Conversion Between Binary, Octal, and Hexadecimal Expansions$_1$

- **Keys**: each octal digit **corresponds to** a block of three binary digits, and each hexadecimal digit **corresponds to** a block of four binary digits.

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Think about how many possibilities are there for four (or three) digits of a binary number. How about 1 digit of a hexadecimal? 1 digit of a octal number?

Conversion Between Binary, Octal, and Hexadecimal Expansions$_2$

- **Example**: Find the octal and hexadecimal expansions of $(11\ 1110\ 1011\ 1100)_2$.
- **Solution**:
  - To convert to octal, we group the digits into blocks of three $(011\ 111\ 010\ 111\ 100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3,7,2,7, and 4. Hence, the solution is $(37274)_8$.
  - To convert to hexadecimal, we group the digits into blocks of four $(0011\ 1110\ 1011\ 1100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3,E,B, and C. Hence, the solution is $(3EBC)_{16}$.

Fractional Binary Numbers **(Optional)**

- **Key**: for binary number, two 1s in the less significant digit equals one 1 in the more significant digit, e.g. the weight for the first digit after dot is 0.5

Binary	**1**	**0**	**1**	**1**	**.**	**1**	**0**	**1**
N-th Digit	4	3	2	1	.	-1	-2	-3
# Smaller (or **Weight**)	$\mathbf{8 = 2^3}$	$\mathbf{4 = 2^2}$	$\mathbf{2 = 2^1}$	$\mathbf{1 = 2^0}$	.	$\mathbf{0.5 = 2^{-1}}$	$\mathbf{0.25 = 2^{-2}}$	$\mathbf{0.125 = 2^{-3}}$
Base 10	$1*2^3+$ $0*2^2+$ $1*2^1+$ $1*2^0+$ $1*2^{-1}+$ $0*2^{-2}+$ $1*2^{-3}$ $=11.625$	$0*2^2+$ $1*2^1+$ $1*2^0+$ $1*2^{-1}+$ $0*2^{-2}+$ $1*2^{-3}$ $=3.625$	$1*2^1+$ $1*2^0+$ $1*2^{-1}+$ $0*2^{-2}+$ $1*2^{-3}$ $=3.625$	$1*2^0+$ $1*2^{-1}+$ $0*2^{-2}+$ $1*2^{-3}$ $=1.625$	.	$1*2^{-1}+$ $0*2^{-2}+$ $1*2^{-3}$ $=0.625$	$0*2^{-2}+$ $1*2^{-3}$ $=0.125$	$1*2^{-3}$ $=0.125$

Algorithms for Binary Integer Operations

- The algorithms for performing operations with integers using their binary expansions are extremely important in computer arithmetic.
- We will describe algorithms for the **addition** and the **multiplication** of two integers expressed in **binary notation**.
  - $a = (a_{n-1}a_{n-2} \dots a_1 a_0)_2$
  - $b = (b_{n-1}b_{n-2} \dots b_1 b_0)_2$
  - Each digit is called a *bit.*

Binary Addition Algorithm$_1$

- Algorithm 2: addition of integers

procedure *add*(a, b: positive integers)
  $c := 0$ {the carry bit from the summation of the last bit: 0 or 1}
  for $j := 0$ to $n - 1$
    $d := \lfloor (a_j + b_j + c)/2 \rfloor$ {carry bit for the summation of the next bit}
    $s_j := a_j + b_j + c - 2d$ {result of current bit}
    $c := d$
  $s_n := c$ {if there is a carry bit at the end}
return($s_n, s_{n-1}, \ldots, s_0$) {the binary expansion of the sum is $(s_n, s_{n-1}, \ldots, s_0)_2$}

- The number of additions of bits used by the algorithm to add two n-bit integers is $O(n)$.

Binary Addition Algorithm$_2$

- **EXAMPLE:** add $a = (1110)_2$ and $b = (1011)_2$.
- **Solution***:* Following the procedure specified in the algorithm, first note that

$$a_0 + b_0 = 0 + 1 = 0 \cdot 2 + 1,$$

so that $c_0 = 0$ and $s_0 = 1$. Then, because

$$a_1 + b_1 + c_0 = 1 + 1 + 0 = 1 \cdot 2 + 0,$$

it follows that $c_1 = 1$ and $s_1 = 0$. Continuing,

$$a_2 + b_2 + c_1 = 1 + 0 + 1 = 1 \cdot 2 + 0,$$

so that $c_2 = 1$ and $s_2 = 0$. Finally, because

$$a_3 + b_3 + c_2 = 1 + 1 + 1 = 1 \cdot 2 + 1,$$

follows that $c_3 = 1$ and $s_3 = 1$. This means that $s_4 = c_3 = 1$. Therefore, $s = a + b = (1\,1001)2$

```
 1 1 1
   1 1 1 0
 + 1 0 1 1
 ---------
 1 1 0 0 1
```

FIGURE 1 Adding $(1110)_2$ and $(1011)_2$.

Binary Multiplication Algorithm$_1$

- Algorithm 3: multiplication of integers

```
procedure multiply(a, b: positive integers)
   for  j := 0 to n − 1
      if b_j = 1 then c_j = a  shifted j places
      else c_j := 0      {c_o,c_1,…, c_n-1 are the partial products}
    p := 0
   for  j := 0 to n − 1
      p  := add(p, c_j)          ←———— O(n)
return p {p is the value of ab}
```

  - The number of additions of bits used by the algorithm to multiply two n-bit integers is $O(n^2)$.

Binary Multiplication Algorithm$_2$

- **EXAMPLE:** Find the product of $a = (110)_2$ and $b = (101)_2$.
- *Solution:* First note that

$$ab_0 \cdot 2^0 = (110)_2 \cdot 1 \cdot 2^0 = (110)_2,$$
$$ab_1 \cdot 2^1 = (110)_2 \cdot 0 \cdot 2^1 = (0000)_2,$$
$$ab_2 \cdot 2^2 = (110)_2 \cdot 1 \cdot 2^2 = (11000)_2.$$

To find the product, add $(110)_2$, $(0000)_2$, and $(11000)_2$. Carrying out these additions (using Algorithm 2, including initial zero bits when necessary) shows that $ab = (1\ 1110)_2$.

$$\begin{array}{r} 110 \\ \times\ 101 \\ \hline 110 \\ 000 \\ 110 \\ \hline 11110 \end{array}$$

FIGURE 2 Multiplying $(110)_2$ and $(101)_2$.

Suggested Exercises

- Exercises 1, 3, 7, and 23 on page 269 in the textbook
  - Solution on page S-25
  - If you are using 7-th edition of this textbook
    - Please find exercise 1, 3, 7, and 23 on page 255

Practice 1

- 1. Convert the decimal expansion of each of these integers to a binary expansion.
  - 231
  - 4532

Practice 2

- 1. Convert the decimal expansion of each of these integers to a binary expansion.
  - 231
    - $231 = 128 + 64 + 32 + 4 + 2 + 1 = (11100111)_2$
  - 4532
    - $4532 = 4096 + 256 + 128 + 32 + 16 + 4 = (1000110110100)_2$

Practice 3

- 2. Convert the binary expansion of each of these integers to a decimal expansion.
  - $(11011)_2$
  - $(1010011)_2$

Practice 4

- 2. Convert the binary expansion of each of these integers to a decimal expansion.
  - $(11011)_2$
    - $(11011)_2 = 2^4 + 2^3 + 2^1 + 2^0 = 16 + 8 + 2 + 1 = 27$
  - $(1010011)_2$
    - $(1010011)_2 = 2^6 + 2^4 + 2^1 + 2^0 = 64 + 16 + 2 + 1 = 83$

Practice 5

- 3. Convert the hexadecimal expansion of each of these integers to a binary expansion.
  - $(80E)_{16}$
  - $(135AB)_{16}$

Practice 6

- 3. Convert the hexadecimal expansion of each of these integers to a binary expansion.
  - $(80E)_{16}$
    - $(1000\ 0000\ 1110)_2$
  - $(135AB)_{16}$
    - $(1\ 0011\ 0101\ 1010\ 1011)_2$

Practice 7

- 4. Find the sum and the product of the pair of numbers. Express your answer as a binary expansion.
  - $(1000111)_2,(1110111)_2$

Practice 8

- 4. Find the sum and the product of the pair of numbers. Express your answer as a binary expansion.
  - $(1000111)_2,(1110111)_2$

```
     1    111
     1000111
+    1110111
---------------
    10111110
```

```
                1000111
*               1110111
-----------------------
                1000111
               1000111
              1000111
             1000111
            1000111
           1000111
-----------------------
        10000100000001
```

Extension 1

- 1. (1) Compute $(1201)_3 \times (12)_3$
  - $(22112)_3$
- (2) Compute the difference

  $(145)_8 - (11011)_2$

  Both numbers are unsigned. Please show the result as a binary number.
  - $(1001010)_2$

Extension 2

- 2. Further readings about binary representations of integers: **one's complement and two's complement** Textbook page 270: question 40 – 53
  - If you are using 7-th edition of this textbook
    - Please find them on page 256

Extension 3

- $(0)_2$-$(1)_2$=? $(0)_2$-$(10)_2$=? $(0)_2$-$(11)_2$=? $(0)_2$-$(100)_2$=?

	0000	**0000**	**0000**	**0000**
	- 1	**- 10**	**- 11**	**-100**
	1111	**1110**	**1101**	**1100**
n	**0001**	**0010**	**0011**	**0100**
Flip	**1110**	**1101**	**1100**	**1011**
+1	**+ 1**	**+ 1**	**+ 1**	**+ 1**
= -n	**1111**	**1110**	**1101**	**1100**

Highest bit is the sign:

0 for positive,

1 for negative

0111……1111 is the max integer

1000……0000 is the min integer

0000……0000 is 0

1111……1111 is -1